

Change Summary

- Project: codexa
- Worktree: isolated session worktree
- Branch: codex/backend/codexa-20260713-codexa-zero-loss-overhead
- Base: main
- Primary implementation commit: 91c88cd
- Verified candidate commit: ea30f6b
- Latest hardening commit: fix(mcp): bound concurrent artifact persistence
- Subject: feat(mcp): reduce agent overhead without capability loss

Primary Implementation Commit Snapshot

91c88cd feat(mcp): reduce agent overhead without capability loss

README.md	147 +-
benchmarks/agent-ab/README.md	71 +-
.../support/mcp-initialize-tools-list-smoke.mjs	188 +++
docs/architecture/codexa-context-server.md	75 +-
docs/architecture/session-memory.md	13 +-
docs/guides/agent-ab.md	240 +++-
docs/guides/new-user-tutorial.md	17 +-
.../codexa-zero-loss-overhead-plan-2026-07-13.md	554 ++++++++
integrations/claude-code/README.md	14 +-
package.json	2 +
plugins/codexa/skills/codexa/SKILL.md	8 +-
scripts/agent-ab-analysis.mjs	1408 ++++++++
scripts/agent-ab.mjs	761 ++++++--
scripts/benchmark-mcp-transport.mjs	495 ++++++
scripts/verify-plugin-package.mjs	4 +-
src/artifacts.ts	11 +-
src/cli.ts	4 +-
src/cli/hooks.ts	46 +-
src/codex-contract.ts	21 +-
src/freshness-authority.ts	12 +
src/init.ts	15 +-
src/mcp.ts	488 +++++-
src/mcp/advanced-mode-kernel.ts	497 ++++++
src/mcp/capability-kernel.ts	106 ++
src/mcp/compaction.ts	160 ++-
src/mcp/decision-guidance.ts	75 ++
src/mcp/decision-kernel.ts	954 ++++++++
src/mcp/decision-policy.ts	107 ++
src/mcp/envelope.ts	66 +-
src/mcp/resources.ts	90 +-
src/mcp/result-artifacts.ts	844 ++++++++
src/mcp/telemetry.ts	340 ++++++
src/mcp/tool-registry.ts	44 +-
src/mcp/tools.ts	304 ++++
src/query/change-plan.ts	139 +-
src/query/post-edit.ts	99 +-
src/query/post-edit/lifecycle.ts	2 +
src/query/post-edit/next-actions.ts	1 -

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src/query/runtime.ts | 61 +-
src/session-memory/runtime.ts | 85 +-
src/task-snapshots.ts | 4 +-
tests/agent-ab-report.test.ts | 3 +-
tests/agent-ab-v2-identity.test.ts | 167 +++
tests/agent-ab-v2.test.ts | 1191 ++++++
tests/benchmark-mcp-transport.test.ts | 93 ++
...jects-malformed-integer-options-instead.test.ts | 34 +-
...s-02-launches-windows-package-local-cmd.test.ts | 8 +-
...exer-06-keeps-planned-post-edit-reviews.test.ts | 6 +-
tests/init.test.ts | 28 +-
tests/mcp-01-keeps-the-primary-mcp-happy.test.ts | 47 +-
tests/mcp-02-does-not-let-stale-codexa.test.ts | 121 +-
...3-records-truncation-metadata-when-post.test.ts | 8 +-
...p-04-reports-package-version-and-codexa.test.ts | 76 +-
tests/mcp-05-suite.test.ts | 190 ++-
...p-06-core-profile-envelopes-never-steer.test.ts | 7 +-
tests/mcp-advanced-auto.test.ts | 233 ++++
tests/mcp-capabilities.test.ts | 240 ++++
tests/mcp-decision-equivalence.test.ts | 417 ++++++
tests/mcp-decision-kernel.test.ts | 409 ++++++
tests/mcp-result-artifacts.test.ts | 362 ++++
tests/mcp-telemetry.test.ts | 340 ++++
...records-bounded-entries-with-provenance.test.ts | 59 +
tests/task-lifecycle.test.ts | 120 +-
63 files changed, 12201 insertions(+), 530 deletions(-)
create mode 100644 benchmarks/agent-ab/support/mcp-initialize-tools-list-smoke.mjs
create mode 100644 docs/plans/codexa-zero-loss-overhead-plan-2026-07-13.md
create mode 100644 scripts/benchmark-mcp-transport.mjs
create mode 100644 src/freshness-authority.ts
create mode 100644 src/mcp/advanced-mode-kernel.ts
create mode 100644 src/mcp/capability-kernel.ts
create mode 100644 src/mcp/decision-guidance.ts
create mode 100644 src/mcp/decision-kernel.ts
create mode 100644 src/mcp/decision-policy.ts
create mode 100644 src/mcp/result-artifacts.ts
create mode 100644 src/mcp/telemetry.ts
create mode 100644 tests/agent-ab-v2-identity.test.ts
create mode 100644 tests/agent-ab-v2.test.ts
create mode 100644 tests/benchmark-mcp-transport.test.ts
create mode 100644 tests/mcp-advanced-auto.test.ts
create mode 100644 tests/mcp-capabilities.test.ts
create mode 100644 tests/mcp-decision-equivalence.test.ts
create mode 100644 tests/mcp-decision-kernel.test.ts
create mode 100644 tests/mcp-result-artifacts.test.ts
create mode 100644 tests/mcp-telemetry.test.ts

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Verification

- git diff --check: passed
- npm run check on ea30f6b: 63 files; 598 passed; 1 explicit opt-in skip; 113 hook smokes
- npm run benchmark:ci on ea30f6b: all hot-path thresholds passed

- npm run eval:ci on ea30f6b: 21 scenarios; score 1; rawRgBetter=0; seed ci-local-ea30f6b74101da7d7c42585d248949db56ed2909
- npm run security:check on ea30f6b: complete check, zero-vulnerability audit, clean public snapshot, package/plugin hygiene, and 25-check installed-package smoke passed
- opt-in pinned v0.12.0 compatibility test on ea30f6b: 2/2 passed using locally built tagged source with pinned commit, lockfile, CLI, and dist-tree identities
- fresh pinned v0.12.0 transport comparison on ea30f6b: 21 to 10 direct schemas; all 21 logical operations advertised; tools/list 59,376 to 29,554 decoded bytes (-50.2%); advertisement plus discovery 59,376 to 42,481 (-28.5%); first task result 55,419 to 16,387 (-70.4%); repeated median 55,419 to 6,372 (-88.5%); four unchanged receipts; 58,595-byte decoded detailed-resource response remained readable
- Codexa post-edit-review: authority complete; all seven invariants satisfied; no drift or unaccounted tests
- Codexa test-plan: passed; local artifact retained outside the repository
- git diff --cached --check: passed
- staged safety scan: passed

The transport comparison measures JSON-serialized decoded MCP application payloads, not wire bytes, tokens, cost, task success, or agent quality. A held-out agent comparison is still required to measure net usefulness and non-inferiority against no Codexa.

PR Hardening

PR CI exposed a same-process retention-lock deadline race. The follow-up uses a bounded per-checkout FIFO so the 500 ms filesystem budget measures foreign process contention, while a separate two-second no-progress budget bounds a stalled local holder. Tests cover a 64-call burst, a 650 ms productive holder, failed-batch recovery, and FIFO substitution without blocking.

Telemetry destinations now require a runner-enforced unique absent path per server session, use exclusive nonblocking creation, and preserve prior evidence. Bounded regular-file reads and nonblocking opens prevent special-file substitution from hanging or causing unbounded reads. Independent adversarial review found no release blocker, capability removal, or domain-specific production behavior.

Notes

Implements decision-safe compact delivery, all 21 logical operations reachable through a 10-schema core surface, bounded lifecycle and artifact persistence, generic loop controls, and schema-v2 agent evaluation. The transport result does not establish zero capability loss or improved agent outcomes; those claims require held-out agent evaluation. Includes a pinned v0.12.0 transport reproducer with explicit claim boundaries.